APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Solar System & Planetary Orbits

SUBJECT: Environmental Studies (EVS101)

CLASS & SECTION: Class 4 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 93 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Create a model or colorful diagram of the eight planets in order with orbital velocities and atmospheric features.

1. Executive Overview & Problem Context

Our Solar System originated approximately 4.6 billion years ago from the gravitational collapse of a giant interstellar molecular gas cloud. Situated in the Orion Cygnus arm of the Milky Way, the system comprises a G-type main-sequence star (The Sun), four inner rocky terrestrial planets, an asteroid belt, four outer Jovian gas/ice giants, and the distant Kuiper Belt.

2. Core Analysis, Solutions & Systematic Working

- **The Central Star: Solar Energy & Gravitation:**

  The Sun accounts for 99.86% of the total mass in the Solar System. Through nuclear fusion in its 15,000,000°C core, hydrogen nuclei fuse into helium, releasing 3.8 × 10² v Watts of radiant energy that governs planetary thermal climates and retains bodies in Keplerian elliptical orbits.

- **Terrestrial Planets (Mercury, Venus, Earth, Mars):**

  Dense, silicate-metallic bodies. Mercury: cratered, atmosphere-less, extreme temperature swings (-180°C to 430°C). Venus: runaway greenhouse effect with 96% CO , atmosphere (surface 465°C). Earth: active hydrologic cycle and nitrogen-oxygen atmosphere sustaining life. Mars: iron oxide dust, thin CO , atmosphere, Olympus Mons volcano.

- **Jovian Giants (Jupiter, Saturn, Uranus, Neptune):**

  Massive hydrogen-helium and volatile ice giants. Jupiter: largest planet, Great Red Spot storm, 95 moons. Saturn: complex ring system composed of billions of water-ice particles. Uranus & Neptune: ice giants enriched in methane giving them vivid cyan and azure hues; Uranus rotates on a unique 98-degree axial tilt.

- **Kepler's Third Law of Planetary Motion:**

  Planetary orbital periods scale with distance from the Sun (T² " r³). Mercury orbits in a swift 88 Earth days at 47.4 km/s, whereas distant Neptune requires 164.8 Earth years at an orbital velocity of 5.4 km/s.

3. Detailed Data Matrix & Systematic Findings

Planet Name	Mean Distance (AU)	Orbital Period	Moons	Distinguishing Feature
Mercury	0.39 AU	88 Earth Days	0	Extreme thermal fluctuation (-180 to 430°C)
Venus	0.72 AU	225 Earth Days	0	Runaway greenhouse atmosphere, retrograde
Earth	1.00 AU	365.25 Days	1 (Moon)	Liquid water oceans & living biosphere
Mars	1.52 AU	687 Earth Days	2	Iron-oxide red soil, Olympus Mons volcano
Jupiter	5.20 AU	11.86 Years	95	Massive magnetosphere, Great Red Spot
Saturn	9.58 AU	29.45 Years	146	Spectacular water-ice planetary ring system
Uranus	19.22 AU	84.02 Years	28	Extreme 98° axial tilt, methane atmosphere
Neptune	30.05 AU	164.79 Years	16	Supersonic winds exceeding 2,100 km/h

4. Student Conclusions & Reflective Synthesis

Planetary science illustrates the unique habitability of Earth within the circumstellar Goldilocks zone. Ongoing space exploration through robotic probes continually deepens our understanding of planetary evolution and potential extraterrestrial biosignatures.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Analytical Depth & Research Methodology: 28 / 30

Problem Solving & Structured Working: 18 / 20

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding astronomical depth! Superb comparative planetary analysis and clear orbital mechanics discussion."

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